TUTOR GUIDE • LEVEL 1 • TERM 1 • LESSON 1 OF 10

Bundles and Singles

Big idea: A number is tens and ones. Ten single sticks make one bundle. A two-digit number tells us how many bundles and how many singles.

90 minutes • live online or face to face • up to 8 children • ages 6-8 • Notice it. Test it. Explain why.

1. At a glance

Strand	Whole Numbers (Numbers to 100: counting a set; Tens and ones (place value); Reading and writing numerals) Strand label for the tutor only. It is not a claim that the program covers the school syllabus, and it must not appear on child or parent pages as 'MOE-aligned'.
Objectives	• Group up to 99 objects into bundles of ten and single ones, and say how many bundles and how many singles. • Read and write a two-digit number from a picture of bundles and singles, and the other way round. • Know that the left digit counts the tens and the right digit counts the ones, and that a 0 in the ones place means no singles. • Show that ten singles can be tied into one more bundle (5 tens 14 ones is 6 tens 4 ones).
Joining cold	**Remember warm-up covers:** Count in tens to 100; Fingers to make 10 (bonds to 10); Ten and some more (11 to 19, then 20s and 30s). Only counting to 20 and showing a number with fingers. Nothing from the holiday course is needed; if the child has done it, skip counting in 10s and bonds to 10 will feel familiar. Lesson 1 is one of the hardest to miss in Term 1 (plan, section 2). A child who misses it watches the recording and does workbook pages 1 and 2 before Lesson 2.
Child materials	• About 50 small things of one kind: paper clips, dried pasta, buttons, lolly sticks or straws cut short • 6 hair ties, rubber bands, small cups or pieces of string (to hold a bundle), or draw rings on paper • The printed workbook (pages 1 to 5) and the kit page (tens bars and ones squares), already cut out by a grown-up • A pencil, a rubber and spare paper
Tutor materials	• A slide deck and a tray of 30 lolly sticks or straws with three rubber bands to show on camera if you wish • A timer slide or a visible clock No ruler is needed. Nothing needs to be bought: everything can be found at home. No matches, toothpicks or anything sharp. Small things never go near the mouth or ears.
Tiers	Start here (6 questions, pages 1 and 2), Think harder (4, page 3), Go further (3, page 4), Investigation and look-back (page 5), kit page (page 6). Labels are neutral; nobody is told which page they are on.
Contest-style move	Used at Think harder only: split a number into tens and ones; digit-clue puzzle (find the number from two clues). Never advertised to children or parents.

Before the lesson

- Solve every workbook question yourself first.
- Tell parents a week before: print workbook pages 1 to 6, cut out the kit page (page 6), find about 50 small things and 6 hair ties or cups.
- Have the slides open and check the timer slide and the kit page are visible on screen share.
- Check the recording settings exclude children's video.

Safeguarding and recording (this lesson)

- Small things stay on the table: nothing goes in the mouth or ears. No matches, toothpicks or anything sharp. Use hair ties, cups or paper rings, not stretched rubber bands that can flick.
- This lesson uses the main room only: no breakout rooms, so the tutor sees every camera. A parent sits nearby but does not give answers.
- Sneak-peek and exit answers are shown on the child's own paper to the camera at the same moment, once. There is no private chat and no board held up in front of the group.
- Never read out a child's name with a right or wrong answer. Never show a child's page in the recording; record the tutor and the screen only, with children's video excluded.
- A child who ticks 'I am still stuck' gets a check-in in the open session next lesson (a quick visit to the shared room or a hand signal, parent nearby). No private messages to children.
- The look-back is never collected in front of the group or read out. A pass is always allowed.

2. Timing plan (90 minutes exactly)

Minutes	Part	Min	Slides	Script fills
0–4	**Opening**	4	3: Bundles and Singles, Our lesson today, Four inquirer rules	3:40
4–10	**Remember warm-up**	6	3: Warm-up: count in tens, Fingers to ten, Ten and some more	4:36
10–14	**Sneak-peek puzzles**	4	1: Sneak-peek puzzles	3:33
14–28	**Explore**	14	3: Explore: grab and bundle, Show me!, What did we notice?	11:13
28–38	**Mini-lesson**	10	8: Ten singles make a bundle, Reading 34, Your turn: write the number, Check: write the number, The zero holds the place, Same digits, different number, Ten singles can tie up, Count: tens, then ones	8:38
38–42	**Brain break**	4	1: Jump the tens, hop the ones	3:28
42–70	**Workbook**	28	5: Workbook page 1: Start here, Check page 1, Workbook page 2: Start here, Check page 2, Choose your next page	25:25
70–76	**Talk time**	6	2: Talk and explain, Back to the sneak-peek	5:12
76–82	**Exit ticket**	6	2: Exit ticket, Check the exit ticket	4:19
82–85	**Look-back**	3	1: Look-back	2:28
85–90	**Flex time and goodbye**	5	1: Well done, inquirers!	0:38
	Total	**90**		**73:10**

The script needs 73:10 of the 90:00 when children are quick. A slow class (speech 10% slower, children and slide changes 20% slower) needs 86:54. The last 5 minutes are a standing buffer: if you are behind, use them; do not start anything new. Clock checks: minute 14 (end of sneak-peek), minute 42 (start of workbook), minute 70 (talk time). If more than 2 minutes behind at minute 42, skip in this order: quick questions in Remember, the last explore number, the quick-fire in the mini-lesson, the optional slide. Never skip the look-back or the camera and privacy moments.

3. Lesson script

Times show when each line should start, counted from the start of the lesson, if you keep to the script. SAY lines are spoken in your own words. REC lines are the short recording-voice cues for children who watch later. Children answer aloud, with fingers or hands; no private chat; no boards held up in front of the group.

0:00 – 4:00 Opening

script 3:40 of 4:00

0:00	SLIDE	Show slide **Bundles and Singles** (s-title)
0:25	REC	*If you are watching the recording: hello! Get a pencil, paper, your workbook and your things to bundle. Pause or rewind any time.*
0:36	SAY	Good morning, inquirers! Cameras on, pencil ready, your things to bundle on the table. This lesson is recorded so that a friend who is away can catch up. We record me and the screen only. You may switch your camera off whenever you like.
0:58	SLIDE	Show slide **Our lesson today** (s-routine)
1:24	SAY	Here is our plan. A warm-up, two puzzles, then we explore with real things. Then I teach you a trick, we stretch, and you work in your workbook.
1:38	SLIDE	Show slide **Four inquirer rules** (s-rules)
2:02	READ	Four inquirer rules. Notice it: look closely. Test it: try it with real things. Explain why: say how you know. Mistakes help us: a guess is fine and you can always pass.
2:20	SAY	A shopkeeper has 47 pencils to count for a customer. Counting one by one is slow, and it is easy to lose count.
2:31	ASK	What could the shopkeeper do to count faster? Tell your screen or your grown-up. [wait about 25 s for answers]
3:03	SAY	Today we learn the shopkeeper's trick: bundles of ten.
3:08	DO	Greet each child by first name and check their camera shows the page and hands. [0:32]
	NOTE	Lesson 1 only: show the routine slide for 20 seconds and name the phases; do not read the table. Tell the group that recordings show only you and the screen.

4:00 – 10:00 Remember warm-up

script 4:36 of 6:00

4:00	SLIDE	Show slide **Warm-up: count in tens** (s-rem1)
4:25	REC	*Recording: join in! Say the numbers out loud and show your fingers. Pause if you need longer.*
4:34	SAY	Stand up if you can. We clap on every number. Count in tens with me: ten, twenty, thirty, all the way to one hundred.
4:46	CHILDREN	Count in tens together, clapping, to 100. Then back from 100 to 0, whispering, shouting on 50. [0:45]
5:30	SLIDE	Show slide **Fingers to ten** (s-rem2)
5:56	SAY	Fingers to ten. I say a number. Show me how many more fingers make ten.
6:03	ASK	Six. How many more to make ten? [wait about 12 s for answers]
6:18	ASK	Three? [wait about 12 s for answers]
6:31	ASK	Eight? [wait about 12 s for answers]
6:44	ASK	Five? [wait about 12 s for answers]
6:56	ASK	Nine? [wait about 12 s for answers]
7:08	SLIDE	Show slide **Ten and some more** (s-rem3)
7:34	SAY	Ten and some more. Look at the full ten-frame and the extra counters.
7:40	ASK	Ten and three more makes? [wait about 12 s for answers]

7:54	**ASK**	Ten and six more? [wait about 12 s for answers]
8:08	**ASK**	Twenty and four more? [wait about 12 s for answers]
8:22	**ASK**	Thirty and seven more? [wait about 12 s for answers]
	IF STUCK	If a child hesitates on 24 or 37, say the tens word slowly (twenty... four) and point to the tens and the ones in the picture. Do not move on until most children have answered; this idea is the lesson.
	NOTE	Answers: 4, 7, 2, 5, 1 then 13, 16, 24, 37. Children answer with fingers, hands or a whispered number. Nobody writes. Joining cold: all you need is counting to 20.

10:00 – 14:00 Sneak-peek puzzles

script 3:33 of 4:00

10:00	**SLIDE**	Show slide **Sneak-peek puzzles** (s-sneak)
10:25	**REC**	*Recording: pause now and try both puzzles on your paper. Guessing is fine. Press play for the next part.*
10:34	**SAY**	Two sneak-peek puzzles: ideas we have not learnt yet. A guess is a great start, and I do not know is welcome. Nobody is marked.
10:47	**READ**	Puzzle one. Count the stars. Write how many. You may draw rings round groups to help.
10:56	**CHILDREN**	Children count the stars and write the number on their own paper. [0:55]
11:51	**READ**	Puzzle two. Mei has 3 bundles of ten sticks and 4 single sticks. Ben has 34 sticks. Could they have the same number? Write yes or no, and why.
12:07	**CHILDREN**	Children write or draw their answer. [0:55]
13:02	**HOLD UP**	Everyone holds their paper to the camera at the same moment, once, after puzzle two. Do not read answers aloud. No private chat.
13:27	**SAY**	Thank you. We will come back to puzzle two in a few minutes.
	NOTE	Puzzle 1 answer: 24 stars. Puzzle 2 answer: yes, 3 tens and 4 ones is 34. Look for good ideas to praise later (a child who drew rings of ten, or who said 3 bundles is 30). Do not write scores or keep a list.

14:00 – 28:00 Explore

script 11:13 of 14:00

14:00	**SLIDE**	Show slide **Explore: grab and bundle** (s-explore1)
14:25	**REC**	*Recording: pause now. Grab a handful, make bundles of ten, then build the numbers I call. Press play when you have tried.*
14:36	**SAY**	Today your hands are the teachers. Grab a handful of your small things, not the whole pile. Make bundles of ten. Count the bundles, then count the singles left over.
14:51	**SAY**	If you finish, try the what-if card: what if you open a bundle?
14:58	**CHILDREN**	Round 1: each child grabs, bundles and counts. Tutor scans the cameras and asks a few children to show their bundles. [3:20]
18:18	**ASK**	Tell me what you have. Say it like this: I have, how many tens, and how many ones. [wait about 50 s for answers]
19:16	**SLIDE**	Show slide **Show me!** (s-explore2)
19:42	**SAY**	Round two: show me! Use your tens bars and ones squares from the kit page. I call a number. You build it, then hold it up to the camera.
19:56	**SAY**	Seventeen.
19:56	**CHILDREN**	Children build 17. Hold up to the camera on a count of three. [0:40]
20:36	**SAY**	Thirty.
20:37	**CHILDREN**	Build 30. [0:35]
21:12	**SAY**	Forty-two.
21:12	**CHILDREN**	Build 42. [0:40]

21:52	SAY	Fifty-eight.
21:53	CHILDREN	Build 58. [0:45]
22:38	SAY	One more. Ravi built thirty-five with 5 bars and 3 squares. Look at your kit. Is Ravi right?
22:47	CHILDREN	Children check with their kit and answer by thumbs up or down, then say why. Answer: no, 5 bars and 3 squares is 53; thirty-five is 3 bars and 5 squares. [0:55]
23:42	SLIDE	Show slide **What did we notice?** (s-explore3)
24:07	ASK	What did you notice? How many singles can be left over? Could you have ten singles left over? Why not? [wait about 45 s for answers]
25:02	SAY	If you have ten singles you can tie them into another bundle. So the singles left over are always nought to nine.
	IF STUCK	A child who counts bundles as ones (3 bundles and 4 singles is 7): hold up a real bundle and say, this one is worth ten. Count it: ten, twenty, thirty, then thirty-one, thirty-two.
	IF STUCK	A child with too few things: use the kit bars and squares for round 1 as well.
	NOTE	Main room only: no breakout rooms in this lesson, so you can see every camera. Pairs are optional: a grown-up may check the building. Building with the kit is for every child; the what-if card is for any child who wants it, and nobody is named.

28:00 – 38:00 Mini-lesson

script 8:38 of 10:00

28:00	SLIDE	Show slide **Ten singles make a bundle** (s-m1)
28:25	REC	*Recording: keep watching and copy the examples. Rewind if you miss a step.*
28:32	SAY	Look. Ten single sticks. When I tie them together I have one bundle. One bundle is worth ten.
28:40	ASK	How many sticks are in one bundle? [wait about 12 s for answers]
28:56	SLIDE	Show slide **Reading 34** (s-m2)
29:21	SAY	Here are 3 bundles and 4 singles. I count the bundles in tens: ten, twenty, thirty. Then I count the singles: thirty-one, thirty-two, thirty-three, thirty-four. Thirty-four sticks.
29:34	SAY	I write it in two boxes. Tens: 3. Ones: 4. That makes 34. The left digit counts the tens. The right digit counts the ones.
29:47	ASK	Which digit tells me how many bundles? [wait about 15 s for answers]
30:06	SLIDE	Show slide **Your turn: write the number** (s-m3)
30:30	READ	Your turn. Write the number for each picture on your paper.
30:36	CHILDREN	Children write three numbers. [0:50]
31:26	SLIDE	Show slide **Check: write the number** (s-m3a)
31:52	READ	Twenty-seven, forty, nineteen.
31:53	SLIDE	Show slide **The zero holds the place** (s-m4)
32:18	SAY	Look at forty. 4 bundles and no singles. We write a zero in the ones box. If we left it out, the 4 would look like 4 ones. The zero holds the place.
32:35	SLIDE	Show slide **Same digits, different number** (s-m5)
33:00	ASK	Two six and six two use the same digits. Which has more sticks, 26 or 62? How do you know? [wait about 30 s for answers]
33:40	SAY	The tens digit is worth more. 62 has 6 bundles. 26 has only 2 bundles. The order of the digits matters.
33:50	SLIDE	Show slide **Ten singles can tie up** (s-m5b)
34:15	SAY	Sometimes you have lots of singles. Here are 2 bundles and 13 singles. Ten of the singles can tie into a new bundle. Now there are 3 bundles and 3 singles. Same sticks, 33 of them.

34:33	**ASK**	Why can we tie up ten singles? What is a bundle worth? [wait about 20 s for answers]
34:59	**SLIDE**	Show slide **Count: tens, then ones** (s-m6)
35:24	**SAY**	Quick-fire. I say a number. You tell me the tens and the ones. Forty-seven.
35:31	**ASK**	Sixty? [wait about 15 s for answers]
35:47	**ASK**	Nineteen? [wait about 15 s for answers]
36:02	**ASK**	Eighty-two? [wait about 15 s for answers]
36:18	**CHILDREN**	Count together: tens, then ones: 10, 20, 30, 31, 32, 33, 34. [0:20]
	NOTE	Key words to use every time: bundle (ten), single (one), tens digit, ones digit. Do not teach 'place value' as a term yet. Do not teach carrying. Keep ten singles making a bundle for the Think harder page.
	IF STUCK	If a child reads 34 as 43: ask them to build it with the kit bars and squares and say the tens first.

38:00 – 42:00 Brain break

script 3:28 of 4:00

38:00	**SLIDE**	Show slide **Jump the tens, hop the ones** (s-break)
38:25	**REC**	*Recording: stand up and join in! Big jump for a ten, small hop for a one.*
38:33	**SAY**	Stand up and get a sip of water. A big jump for every ten, a small hop for every one. Thirty-four is three big jumps and four small hops.
38:48	**CHILDREN**	Call 12, 25, 40, 31. Children jump the tens and hop the ones. Count the jumps and hops aloud together. [2:40]

42:00 – 70:00 Workbook

script 25:25 of 28:00

42:00	**SLIDE**	Show slide **Workbook page 1: Start here** (s-ws1)
42:25	**REC**	*Recording: pause now and do workbook page 1. When you finish a page, press play to check it.*
42:34	**READ**	Page 1, Start here. Question 1: count the sticks and fill in the bundles, the singles and the number. Question 2: draw 3 bundles and 5 singles. Question 3: write the numbers.
42:51	**CHILDREN**	Page 1, questions 1 to 3. Tutor walks the cameras, asks how do you know. Wait 10 seconds before helping. [9:00]
51:51	**SLIDE**	Show slide **Check page 1** (s-ws1ans)
52:16	**READ**	Twenty-six. Thirty-five. Forty-two, seventy, nineteen, eighty-five. Mark your own page in another colour.
52:24	**SAY**	After question 3 you may choose: carry on, or try Think harder. Everyone may choose, at any time.
52:33	**SLIDE**	Show slide **Workbook page 2: Start here** (s-ws2)
52:58	**READ**	Page 2. Question 4: fill in tens and ones. Question 5: Ali has 47 straws. Question 6: Siti wrote 43. Is she right?
53:10	**CHILDREN**	Page 2, questions 4 to 6. [8:00]
61:10	**SLIDE**	Show slide **Check page 2** (s-ws2ans)
61:35	**READ**	Five tens and eight ones. Four tens, no ones. Seven tens and one one. Four bundles, seven left over. Siti should write thirty-four.
61:48	**SLIDE**	Show slide **Choose your next page** (s-choice)
62:13	**SAY**	When you have finished Start here, you can choose Think harder, Go further or the Investigation. Everyone may choose. You can swap at any time.
62:25	**CHILDREN**	Choice time: Think harder (page 3), Go further (page 4), the Investigation (page 5), or finish and check your own work. Tutor never says who is on which page. [5:00]
	IF STUCK	Q2 (drawing): draw a bundle as ten short lines with a ring round them. Q4 (zero): ask how many singles are left over in 40. Q6: build 3 bundles and 4 singles with the kit, say the tens first.

	IF STUCK	A child who is still stuck ticks 'I am still stuck' on the look-back page. You check in during the open session in the next lesson (a quick visit to the shared room or a hand signal, parent nearby). Never a private message.
	NOTE	Think harder is for ideas taught today in a new setting. Go further is for mastery and extension. Do not announce who is on which page. Pairs are never split unless both children agree.

70:00 – 76:00 Talk time

script 5:12 of 6:00

70:00	SLIDE	Show slide **Talk and explain** (s-talk)
70:25	REC	*Recording: pause. Tell a grown-up your answer to the talk question, and say why.*
70:32	READ	Mei says 26 and 62 are the same, because they use the same digits. Do you agree? Use bundles to explain.
70:43	CHILDREN	Build 26 and 62 with your kit bars and squares. Which one has more? [0:50]
71:33	CHILDREN	Talk to the grown-up next to you or to your screen for one minute. Use the sentence starter: I think... because... [1:00]
72:33	ASK	Who would like to share? Volunteers first. A pass is always fine. [wait about 90 s for answers]
74:09	SLIDE	Show slide **Back to the sneak-peek** (s-sneakans)
74:34	SAY	Remember puzzle two? Mei has 3 bundles and 4 singles. Ben has 34 sticks. Yes! They have the same number, thirty-four.
74:45	ASK	Did anyone change their mind about puzzle two while we worked? You can just nod. [wait about 20 s for answers]
	NOTE	Model answer: No. 26 is 2 bundles and 6 singles. 62 is 6 bundles and 2 singles. 62 has more sticks. Volunteers first; never call a child by name to answer; no tallies, no rankings. Praise the reason, not the speed.

76:00 – 82:00 Exit ticket

script 4:19 of 6:00

76:00	SLIDE	Show slide **Exit ticket** (s-exit)
76:25	REC	*Recording: pause and do the exit ticket on your own. Then press play to check it.*
76:33	SAY	Exit ticket. Three questions, on your own, without looking back. Write your answers on page 5.
76:41	CHILDREN	Children answer E1 to E3 on their own. [2:30]
79:11	HOLD UP	Everyone holds their ticket to the camera at the same moment so only you see it. No private chat.
79:36	SLIDE	Show slide **Check the exit ticket** (s-exitans)
80:01	READ	Forty-eight. Three tens and seven ones. Seventy, and the zero is needed to show there are no ones.
80:11	SAY	Check your own ticket in a different colour. Well done. I will stamp your inquiry journal.
	NOTE	Mark S, D or H in the tutor tracker (tutor only): S = secure on all three, D = developing (one wrong), H = needs help (two or three wrong). If D or H, plan two minutes of catch-up at the start of Lesson 2, in the open session, no names. Do not read out any child's result.

82:00 – 85:00 Look-back

script 2:28 of 3:00

82:00	SLIDE	Show slide **Look-back** (s-look)
82:25	REC	*Recording: pause and fill in your look-back box. You can pass.*
82:30	SAY	Look-back. Tick what was hard and what helped you. Write or draw one thing you did well. If you like, tick one thing to try next time. You can pass. Nobody reads it out.
82:48	CHILDREN	Children fill in the look-back box on page 5. [1:40]

85:00 – 90:00 Flex time and goodbye

script 0:38 of 5:00

85:00	**SLIDE**	Show slide **Well done, inquirers!** (s-end)
85:25	**SAY**	Well done, inquirers. Next lesson is Number Words and the Number Line. You need only a pencil and paper. Goodbye, and thank you to the grown-ups.
	SLIDE	Show slide **Go further: what if?** (s-opt) *(optional)*
	SAY	Optional, only if you are on time: what is the biggest number you can make with 9 bundles and 9 singles? And what happens if you add one more single?
	CHILDREN	Short think time, then take two answers. [1:30]
	NOTE	This is the standing buffer. If you are behind, finish the look-back and say goodbye. Do not start anything new. Optional slide answer: 99; one more is ten bundles, which is called a hundred (next lessons).

4. Questions, answers, hints and common mistakes

Sneak-peek puzzles (minutes 10 to 14)

Puzzle 1 (access): Count the stars. Write how many. You may draw rings round groups to help.

> **Answer: 24**

Puzzle 2 (think): Mei has 3 bundles of ten sticks and 4 single sticks. Ben has 34 sticks. Could they have the same number of sticks? Write yes or no, and why.

> **Answer: Yes. 3 bundles of ten is 30, and 4 more is 34.**

Unmarked. Papers are shown to the camera once, together. Reveal puzzle 2 in Talk time.

Start here (workbook pages 1 and 2)

S1 **Count the sticks.**



a) ___ bundles of ten
b) ___ single sticks
c) ___ sticks altogether

> **Answer: 2 bundles, 6 singles, 26 sticks**

Hint (say this first): Count the bundles in tens: 10, 20. Then count the singles on: 21, 22, 23, 24, 25, 26.
Worked solution: Count the 2 bundles in tens: 10, 20. Then count on the 6 singles: 21, 22, 23, 24, 25, 26. So 2 bundles, 6 singles, 26 sticks.
Common mistakes: Counting each bundle as 1 (2 + 6 = 8).; Writing the digits the wrong way round (62).
Time for a middle child: about 1.5 min.

S2 **Draw 3 bundles of ten and 5 single sticks in the box. Then write the number.**

The number is ___

> **Answer: 35 (3 bundles drawn, each with ten sticks, and 5 singles)**

Hint (say this first): Draw a bundle as ten short lines with a ring round them. Draw 3 of those, then 5 more short lines on their own.
Worked solution: Draw 3 rings each holding ten short lines (that is 30), then 5 single lines. Count: 10, 20, 30, 31, 32, 33, 34, 35.
Common mistakes: Drawing 3 sticks and 5 sticks instead of 3 bundles of ten.; Writing 53.
Time for a middle child: about 3 min.

S3 **Write the number.**

a) 4 tens and 2 ones = ___
b) 7 tens and 0 ones = ___
c) 1 ten and 9 ones = ___
d) 5 ones and 8 tens = ___

> **Answer: 42, 70, 19, 85**

Hint (say this first): Say the tens first, then the ones. In (d) the ones come first in the words, but the tens digit still goes on the left.
Worked solution: (a) 4 tens is 40, and 2 more is 42. (b) 7 tens is 70 and there are no ones: 70. (c) 10 and 9 is 19. (d) 8 tens is 80, and 5 ones makes 85.
Common mistakes: Writing 7 for 7 tens and 0 ones.; (d) writing 58 because 5 was said first.
Time for a middle child: about 3 min.

S4 **Fill in the tens and ones.**

a) 58 = ___ tens and ___ ones
b) 40 = ___ tens and ___ ones
c) 71 = ___ tens and ___ ones

Answer: 5 tens 8 ones; 4 tens 0 ones; 7 tens 1 one

Hint (say this first): The left digit counts the tens. The right digit counts the ones. You can build it with your kit.
Worked solution: (a) 58: the 5 is the tens digit, the 8 is the ones digit. (b) 40: 4 tens and no ones. (c) 71: 7 tens and 1 one.
Common mistakes: Saying 40 has 4 ones.; Swapping the two digits.
Time for a middle child: about 2.5 min.

S5 **Ali has 47 straws. He ties them into bundles of ten.**

a) How many full bundles can he make? ___
b) How many straws are left over? ___

Answer: 4 bundles, 7 left over

Hint (say this first): 47 is 4 tens and 7 ones. Draw it if you like.
Worked solution: 47 is 4 tens and 7 ones, so Ali makes 4 full bundles and has 7 straws left over (4 bundles hold 40 straws).
Common mistakes: Answering 47 for the bundles.; Saying 3 bundles and 17 left over: 17 singles can still make another bundle.
Time for a middle child: about 2.5 min.

S6 **Siti has 3 bundles of ten and 4 single sticks. She writes 43. Is Siti right? Write the number she should write, and say what she mixed up.**

She should write ___

Answer: No. 34. She wrote the tens and ones digits the wrong way round (43 is 4 tens and 3 ones).

Hint (say this first): Build 3 bundles and 4 singles with your kit. Say the tens first.
Worked solution: 3 bundles is 30 and 4 singles makes 34. Siti wrote 43, which is 4 tens and 3 ones. The tens digit goes on the left.
Common mistakes: Saying she is right because the digits 3 and 4 are there.; Not saying what she mixed up (any clear sentence about the digits is fine).
Time for a middle child: about 2 min.

Think harder (page 3): the taught idea in a new setting

T1 **Mei has 5 bundles of ten and 14 loose sticks. How many sticks has she? Hint: can she make another bundle?**

Mei has ___ sticks

Answer: 64 (14 loose sticks make 1 more bundle and 4 left, so 6 tens and 4 ones)

Hint (say this first): Take 10 of the loose sticks and tie them into a bundle. How many bundles now? How many singles?
Worked solution: 14 loose sticks = 1 bundle and 4 singles. So Mei has 5 + 1 = 6 bundles and 4 singles: 64 sticks. Check: 50 + 14 = 64.
Common mistakes: Writing 514 by joining the digits.; Writing 5 tens and 14 ones as 5 + 14 = 19.
Time for a middle child: about 4 min.

T2 **41 is 4 tens and 1 one. Open one bundle. Now 41 is 3 tens and ___ ones.**

3 tens and ___ ones

Answer: 11 (3 tens is 30, and 30 + 11 = 41)

Hint (say this first): A bundle has ten sticks. If you open one, those ten sticks become singles, and you still have the 1 you had.
Worked solution: Opening a bundle gives 10 more singles. 1 single + 10 singles = 11 singles, and 3 bundles are left. 30 + 11 = 41, so it is the same number.
Common mistakes: Writing 10 (forgetting the 1 single already there).; Writing 1 (forgetting the opened bundle).
Time for a middle child: about 3 min.

T3 **Mystery number. Clue 1: I am between 30 and 50. Clue 2: my ones digit is 6. Clue 3: my number of tens is even. What number am I?**

I am ___

Answer: 46

Hint (say this first): List the numbers between 30 and 50 that end in 6. Then check the tens.
Worked solution: Between 30 and 50 and ending in 6: 36 and 46. 36 has 3 tens (odd). 46 has 4 tens (even). So the number is 46.
Common mistakes: Stopping at 36 and not checking clue 3.; Listing numbers that start with 6.
Time for a middle child: about 5 min.

T4 **Mei has 7 tens and 2 ones. Ali has 6 tens and 12 ones. Who has more, or do they have the same? Show how you know.**

Answer: The same: 72 each (6 tens 12 ones is 7 tens 2 ones)

Hint (say this first): Take 10 of Ali's 12 loose ones and tie them into a bundle. How many bundles and ones does Ali have now?
Worked solution: Ali's 12 ones make 1 more bundle and 2 ones. So Ali has 7 tens and 2 ones, the same as Mei. Both have 72.
Common mistakes: Saying Ali has more because 12 is bigger than 2.; Adding 6 + 12 = 18 and comparing 18 with 9.
Time for a middle child: about 5 min.

Go further (page 4): mastery and extension

G1 **Ben has 6 bundles of ten and 14 loose sticks. Siti has 85 loose sticks. Who has more sticks? How many more?**

Answer: Siti. Ben has 74, Siti has 85, so Siti has 11 more.

Hint (say this first): Work out Ben's number first. Remember 14 loose sticks make a bundle and 4 more.
Worked solution: Ben: 6 tens and 14 ones is 7 tens and 4 ones, so 74. Siti: 85. 85 is 8 tens 5 ones and 74 is 7 tens 4 ones: 1 ten and 1 one more, which is 11 more.
Common mistakes: Comparing only 6 and 85 without working out Ben's whole number.; Not opening 14 ones into a bundle before comparing.
Time for a middle child: about 5 min.

G2 **A shop sells pencils in boxes of ten. Mei needs 60 pencils. She has 4 full boxes and 18 loose pencils. How many more pencils does she need?**

Answer: 2 more pencils (4 boxes and 18 pencils is 58)

Hint (say this first): First find how many pencils she has now. Remember that 18 loose pencils make 1 more box and 8 pencils.
Worked solution: 4 boxes is 40 pencils. 40 + 18 = 58. She needs 60, so she needs 60 - 58 = 2 more.
Common mistakes: Saying 18 more (the loose pencils).; Forgetting the 4 full boxes hold 40 pencils.
Time for a middle child: about 5 min.

G3 **How many different ways can you show 30 with bundles of ten and loose sticks, if you may open bundles? Write each way as ___ tens and ___ ones. (3 tens and 0 ones is one way.)**

Answer: 4 ways: 3 tens 0 ones; 2 tens 10 ones; 1 ten 20 ones; 0 tens 30 ones

Hint (say this first): Start with 3 tens. Open one bundle. What do you have now? Keep going until there are no bundles.
Worked solution: 3 tens 0 ones. Open one bundle: 2 tens 10 ones. Open another: 1 ten 20 ones. Open the last: 0 tens 30 ones. That is 4 ways. Each time one ten becomes 10 ones.
Common mistakes: Stopping at 3 tens 0 ones and 0 tens 30 ones (2 ways).; Counting 2 tens 10 ones twice.
Time for a middle child: about 6 min.

Investigation (page 5): Swap the digits

Easy first step: Write 14 and 41. Which number has more sticks? Then try 26 and 62, 52 and 25, and 77.
Then: Pick a two-digit number. Swap its two digits. Is the new number bigger, smaller or the same? Is it always one of these three? When does each happen?

> **Model answer: Three things can happen. If the tens digit is smaller than the ones digit (14, 26), swapping makes a bigger number (41, 62). If the tens digit is bigger (52, 91), swapping makes a smaller number (25, 19). If the two digits are the same (77), the number stays the same.**

Look for: A child who tests several numbers, says what they notice and gives a reason using tens and ones. A number that ends in 0 (30) becomes 03, which is just 3, so it is smaller: this fits 'tens digit bigger'.
Common mistakes: Saying swapping always makes the number bigger (testing only 14 and 26).; Not testing a number with equal digits.

Talk and explain (minutes 70 to 76)

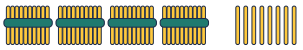
Mei says 26 and 62 are the same, because they use the same digits. Do you agree? Use bundles to explain. Sentence starter: I think ... because ...

> **Model answer: No. 26 is 2 bundles of ten and 6 singles. 62 is 6 bundles of ten and 2 singles. 62 has more sticks, because the tens digit counts bundles of ten.**

Look for: A reason that uses tens and ones (not just 'because 6 is bigger'). Volunteers first, a pass is always fine, never call a child by name to answer, no tallies.

Exit ticket (minutes 76 to 82): 3 questions

E1 **How many sticks?**



___ sticks

> **Answer: 48**

Worked solution: 4 bundles is 40, and 8 singles makes 48.
Common mistakes: Writing 84.; Writing 12 (4 + 8).

E2 **Fill in the tens and ones.**

37 = ___ tens and ___ ones

> **Answer: 3 tens and 7 ones**

Worked solution: The left digit 3 counts the tens, the right digit 7 counts the ones.
Common mistakes: Swapping: 7 tens and 3 ones.

E3 **Write the number: 7 tens and 0 ones. Why do we need the 0?**

The number is ___

> **Answer: 70. The 0 shows there are no ones, and it keeps the 7 in the tens place (without it, it would look like 7 ones).**

Worked solution: 7 bundles of ten and no singles is 70. The zero holds the ones place.
Common mistakes: Writing 7.; Writing 700 or 7 0 as 7 + 0.

Look-back (minutes 82 to 85)

Children tick or write: *Something I did well*; *The hardest part was* (knowing which digit is the tens; making bundles of ten; the zero in 40 or 70; opening a bundle (Think harder); something else); *What helped me* (bundles or objects; drawing it; trying again; I asked for help; I was not stuck); *Next time I will try* (drawing it; using objects; checking my answer; asking a question); *I am still stuck (my tutor will come and help)* and *I would like to pass*. Never collected or read out in front of the group. A child who ticks still stuck gets a check-in in the open session next lesson.

5. Home practice and online practice

#	Tier	Question	Answer
H1	Start here	A box has 3 bundles of ten pencils and 8 loose pencils. How many pencils are there?	38
H2	Start here	Fill in: 65 = ___ tens and ___ ones.	6 tens and 5 ones
H3	Start here	Write the number: 9 ones and 2 tens.	29
H4	Think harder	Ben has 4 tens and 12 ones. Is that 52 or 412? Show how you know.	52 (12 ones make 1 ten and 2 ones, so 5 tens and 2 ones)
H5	Go further	At home: find 30 to 50 small things (pasta, clips, buttons). Make bundles of ten with bands, cups or rings on paper. Write: ___ tens and ___ ones = ___ . Then open one bundle and write the number a different way.	Any: for example 3 tens and 7 ones = 37, and also 2 tens and 17 ones = 37. Check: tens x 10 + ones = how many things.

Online practice (`out/l01/practice`, key `t1l01`): 14 items; 10 have a template that makes unlimited similar questions with a computed answer. Every template variant is checked by the validator against a brute-force oracle and the runner's own JavaScript. No timers, no scores, no streaks.

6. Parent line for the term report

Your child is learning to see numbers as tens and ones and to say why. This week: reading and writing two-digit numbers from bundles of ten and single ones.

Term line (plan, section 5): Your child is learning to see numbers as tens and ones, to notice patterns and to say why. Add one named strength and one next step from your own observation. No marks, no ranking, no claim of exam results.

7. Slide list

#	Time	Slide	On screen	Speaker note
1	0–4	**Bundles and Singles** s-title	TODAY WE WILL: / Make bundles of ten, and use them to read and write numbers.	Lesson 1 of Term 1, 90 minutes. Objective: read and write two-digit numbers as bundles of ten and single ones. Time plan: opening 0 to 4, warm-up 4 to 10, sneak-peek 10 to 14, explore 14 to 28, mini-lesson 28 to 38, brain break 38 to 42, workbook 42 to 70, talk 70 to 76, exit ticket 76 to 82, look-back 82 to 85, flex time and goodbye 85 to 90 (the buffer: do not start anything new). Think harder, Go further and the Investigation are for every child to choose. Nobody is told which page they should be on. Tell the group the lesson is recorded (you and the screen only) and that a child may keep the camera off.
2	0–4	**Our lesson today** s-routine	First | What we do / Remember | Count in tens and show ten with fingers / Sneak-peek | Two puzzles. A guess is fine. / Explore | Bundle real things / Learn | Read and write numbers / Move | Jump and hop / Workbook | Start here, then you choose / Talk | Explain your idea / Exit and look-back | What did I learn?	Show this for about 20 seconds and name the phases only; do not read the table. Lesson 1 only: it introduces the routine every lesson follows.
3	0–4	**Four inquirer rules** s-rules	Notice it. Look closely. / Test it. Try it with real things. / Explain why. Say how you know. / Mistakes help us. A guess is fine. You can pass. / RECORDING: / We record the tutor and the screen only. You may keep your camera off.	Read the four rules together. Say the recording notice in your own words: the platform records the tutor and the shared screen; children's video is excluded; names are kept off the screen. A parent or child who does not wish to be in the recording can keep the camera off. Then tell the shopkeeper story (47 pencils) and ask the group what a shopkeeper could do.
4	4–10	**Warm-up: count in tens** s-rem1	Clap on every number / 10 20 30 40 50 60 70 80 90 100 / Now count back from 100. Whisper, and shout on 50!	Count in tens to 100 with a clap on each number, then back to 0. This is a spiral review of skip counting in tens (holiday topic) and lets a new child join: they hear it before they say it.
5	4–10	**Fingers to ten** s-rem2	[picture: ten_frame] / I show 6. How many more fingers make 10? / 6 3 8 5 9	Say each number; children show on their fingers how many more make ten. Answers: 6 needs 4, 3 needs 7, 8 needs 2, 5 needs 5, 9 needs 1. Hands up on camera is fine. No writing, no marks.
6	4–10	**Ten and some more** s-rem3	[picture: ten_frame] / 10 and 3 more is ___ / 10 + 6 20 + 4 30 + 7	Answers: 13, 16, 24, 37. Say the tens word slowly, then the ones: twenty... four. This is the bridge to bundles of ten and singles. Show 20 and 30 as two or three full ten-frames if a child is unsure.
7	10–14	**Sneak-peek puzzles** s-sneak	Ideas we have not learnt yet. A guess is a great start. Nobody is marked. / PUZZLE 1: anyone can try: / Count the stars. Write how many. / You may draw rings round groups. / [picture: scatter] / PUZZLE 2: think: / Mei has 3 bundles of ten sticks and 4 single sticks. / Ben has 34 sticks. / Could they have the same number? Write yes or no, and why.	Puzzle 1 (24 stars) is an access question: any child can try it by counting or drawing rings. Puzzle 2 is a thinking question on the lesson's key idea. Say: two sneak-peek puzzles, ideas we have not learnt yet; a guess is a great start and I do not know is welcome; nobody is marked. Children write on their own paper. Everyone holds the paper to the camera once, at the same moment, after puzzle 2; there is no private chat and no board in front of the group. Answers: 24; yes (3 tens and 4 ones is 34). Do not reveal Puzzle 2 until Talk time. Do not write scores or keep a list of who found it easy.

8	14–28	**Explore: grab and bundle** s-explore1	Grab a handful of small things. Not the whole pile. / Make bundles of ten: a band, a cup, or a ring on paper. / Count the bundles. Count the singles left over. / Say: I have ___ tens and ___ ones. / WHAT IF?: / What if you open a bundle? Can you show your number a different way?	Round 1, about 4 minutes. Objects first, no writing yet. Children bundle in tens and count the singles. Ask a few children to show their bundles to the camera. The what-if card is available from the start so nobody waits: it previews the Think harder idea (opening a bundle), and no child is named for choosing it. A child with too few things can use the kit bars and squares.
9	14–28	**Show me!** s-explore2	Build each number with your tens bars and ones squares. Hold it up to the camera when I count to three. / [picture: bundles 2 tens 3 ones] / 17 30 42 58	Round 2, about 4 minutes, main room. Call each number aloud and show it on the slide. Children build with the kit bars (tens) and squares (ones). On a count of three, everyone holds their build to the camera. Check the zero case (30) and the larger ones (42, 58). Answers: 17 is 1 bar and 7 squares; 30 is 3 bars; 42 is 4 bars and 2 squares; 58 is 5 bars and 8 squares.
10	14–28	**What did we notice?** s-explore3	How many singles can be left over? / Could you have ten singles left over? Why not? / Is it quicker to count 40 singles, or 4 tens?	Take one or two answers from volunteers. Key idea: the singles left over are always 0 to 9, because ten singles tie into another bundle. Counting in tens is quicker and easier to keep track of. Repeat a child's answer in your own words so the recording picks it up.
11	28–38	**Ten singles make a bundle** s-m1	[picture: ten_to_bundle] / 10 singles = 1 bundle of ten	Show ten single sticks, then tie them into one bundle. A bundle is worth ten. Ask: how many sticks are in one bundle? Answer: ten.
12	28–38	**Reading 34** s-m2	[picture: bundles 3 tens 4 ones] / 30 and 4 more is 34. The left digit counts the tens. The right digit counts the ones.	Think aloud: count the bundles in tens (10, 20, 30) then the singles (31, 32, 33, 34). Write 3 in the tens box and 4 in the ones box. Ask: which digit tells me how many bundles? Answer: the left one, 3.
13	28–38	**Your turn: write the number** s-m3	Write on your paper. Timer: 1 minute / [picture: bundles 2 tens 7 ones] / ___ / [picture: bundles 4 tens 0 ones] / ___ / [picture: bundles 1 tens 9 ones] / ___	Each child writes on their own paper. Walk the cameras. Do not ask children to hold boards up in front of the group.
14	28–38	**Check: write the number** s-m3a	[picture: bundles 2 tens 7 ones] / 27 / [picture: bundles 4 tens 0 ones] / 40 / [picture: bundles 1 tens 9 ones] / 19	Answers: 27, 40, 19. Read each answer aloud so a child watching the recording can mark their own work.
15	28–38	**The zero holds the place** s-m4	[picture: bundles 4 tens 0 ones] / No singles, so the ones box has a 0. Without the 0, 4 would look like 4 ones.	Zero ones means no singles. The zero is needed so the 4 stays in the tens place. A common slip is to write 4 for 4 tens.
16	28–38	**Same digits, different number** s-m5	[picture: bundles 2 tens 6 ones] / [picture: bundles 6 tens 2 ones] / Which has more sticks? How do you know?	26 is 2 bundles and 6 singles. 62 is 6 bundles and 2 singles. 62 has more because the tens digit is worth more. The order of the digits matters. This sets up the Talk time question.
17	28–38	**Ten singles can tie up** s-m5b	[picture: bundles 2 tens 13 ones] / [picture: bundles 3 tens 3 ones] / 13 ones = 1 ten and 3 ones	Show 2 bundles and 13 singles, then tie ten of the singles into a third bundle: 3 bundles and 3 singles. Both pictures are 33 sticks. This is the idea Think harder uses (5 tens and 14 ones). Keep it to about a minute: it is a worked example, not a new skill to drill.
18	28–38	**Count: tens, then ones** s-m6	10 20 30 31 32 33 34 / Say it together	Count the 3 bundles then the 4 singles together. This is the same count as the explore. Keep it short.

19	38–42	**Jump the tens, hop the ones** s-break	A big jump for every ten. A small hop for every one. / 12 25 40 31 / 34 is 3 big jumps and 4 small hops. Sip some water.	Four minutes. Stand up on camera. Call each number and have children jump the tens and hop the ones; count the jumps and hops aloud. No equipment. Keep it gentle for small spaces.
20	42–70	**Workbook page 1: Start here** s-ws1	Questions 1 to 3. Timer: 8 minutes / questions S1, S2, S3	Read questions 1 and 2 aloud. Children work on their own; the tutor walks the cameras and asks how do you know. Wait 10 seconds before helping. After question 3 any child may choose to carry on or try Think harder.
21	42–70	**Check page 1** s-ws1ans	answers S1, S2, S3	Read each answer aloud. Children mark their own page in a different colour. Never ask a child to show a wrong answer to the group.
22	42–70	**Workbook page 2: Start here** s-ws2	Questions 4 to 6. Timer: 7 minutes / questions S4, S5, S6	Read question 6 aloud: it is a spot-the-mistake. Children may draw bundles to decide.
23	42–70	**Check page 2** s-ws2ans	answers S4, S5, S6	Read each answer aloud so a child watching the recording can mark their own work. Question 6 answer: no, Siti should write 34, because 43 is 4 tens and 3 ones. Never ask a child to show a wrong answer to the group.
24	42–70	**Choose your next page** s-choice	Page | What it is / Think harder (page 3) | Our idea in a new setting / Go further (page 4) | Bigger puzzles / Investigation (page 5) | Always, sometimes or never? / Check | Finish and check your own work / Everyone may choose. You can swap at any time.	Invite the whole group at once, for example: when you have finished Start here, you can choose Think harder, Go further or the Investigation. Never announce who is on which page and never name a child. Pairs are not split unless both children agree. Model answers are in the tutor guide. About 4 minutes.
25	70–76	**Talk and explain** s-talk	MEI SAYS: / 26 and 62 are the same, because they use the same digits. / Do you agree? Use bundles to explain. / I think … because …	One minute of talk to the grown-up next to you, or to your screen. Then volunteers first; a pass is always fine; never call a child by name to answer. Model answer: no. 26 is 2 bundles and 6 singles. 62 is 6 bundles and 2 singles, so 62 has more. Repeat each child's answer in your own words for the recording.
26	70–76	**Back to the sneak-peek** s-sneakans	[picture: bundles 3 tens 4 ones] / Yes! Mei and Ben can have the same number: 34.	Reveal puzzle 2 now. Say: many of you wrote a good reason. Do not read out anyone's earlier answer.
27	76–82	**Exit ticket** s-exit	On your own. No looking back. Timer: 3 minutes / questions E1, E2, E3	Children answer on page 5. After the timer everyone holds the ticket to the camera at the same moment, so only the tutor sees it (no private chat). Stamp the inquiry journal.
28	76–82	**Check the exit ticket** s-exitans	answers E1, E2, E3	Read each answer aloud. Children check their own ticket in a different colour.
29	82–85	**Look-back** s-look	[look-back box]	Three minutes. Children tick or write the look-back box on page 5. A pass is always allowed. It is never marked, never collected in front of the group and never read out. If a child ticks I am still stuck, check in during the open session next lesson (a quick visit to the shared room or a hand signal, parent nearby); never a direct message. The Next time I will try row is optional; at the start of the next lesson you may say to the whole group: today you could try drawing it, using objects or checking your answer.
30	85–90	**Well done, inquirers!** s-end	Show a grown-up one thing you found out today. / NEXT LESSON: / Number Words and the Number Line / You need a pencil and paper.	Say goodbye and thank the grown-ups. Minutes 85 to 90 are flex time: the buffer. If you ran late, finish any open item (stamps, the look-back). Do not start anything new.

31	optional	**Go further: what if?** s-opt	What is the biggest number you can make with 9 bundles and 9 singles? / What happens if you add one more single? / [picture: bundles 9 tens 9 ones]	Optional, only if you are on time. Answer: 99. One more single makes 10 singles, which tie into a tenth bundle: ten bundles. Ten tens have a name: a hundred (Lesson 2 onwards). Take two answers and move on.

8. Recording cues and chapter markers

Say each cue once, to the camera, at the start of the part. Add a chapter marker at each time. A child watching alone cannot ask you a question, so say the page and question number aloud, read each question and each answer aloud, and repeat a child's answer in your own words.

Chapter	Part	You say (recording voice)	A child watching later
0:00	Opening	*If you are watching the recording: hello! Get a pencil, paper, your workbook and your things to bundle. Pause or rewind any time.*	Gather your workbook, pencil, paper and things to bundle. Press play when ready.
4:00	Remember warm-up	*Recording: join in! Say the numbers out loud and show your fingers. Pause if you need longer.*	Play along. Say and show each answer before the video does.
10:00	Sneak-peek puzzles	*Recording: pause now and try both puzzles on your paper. Guessing is fine. Press play for the next part.*	Pause. Try both puzzles on your own paper. A guess is fine.
14:00	Explore	*Recording: pause now. Grab a handful, make bundles of ten, then build the numbers I call. Press play when you have tried.*	Pause. Grab and bundle, then build each number with your kit. Write what you find.
28:00	Mini-lesson	*Recording: keep watching and copy the examples. Rewind if you miss a step.*	Watch and copy the examples. Rewind any step you missed.
38:00	Brain break	*Recording: stand up and join in! Big jump for a ten, small hop for a one.*	Stand up and jump the tens and hop the ones.
42:00	Workbook	*Recording: pause now and do workbook page 1. When you finish a page, press play to check it.*	Pause. Do page 1, then play to check. Then page 2. Then choose a page to try.
70:00	Talk time	*Recording: pause. Tell a grown-up your answer to the talk question, and say why.*	Pause. Explain your answer to someone at home, with a reason.
76:00	Exit ticket	*Recording: pause and do the exit ticket on your own. Then press play to check it.*	Pause. Do the exit ticket alone, then play to check it.
82:00	Look-back	*Recording: pause and fill in your look-back box. You can pass.*	Pause. Fill in the look-back box, or pass.
85:00	Flex time and goodbye	*Recording: well done for finishing! Show a grown-up one thing you found out.*	Show a grown-up your workbook and tell them one thing you found out.

9. Timing check (model, not a stopwatch)

Part	Allocated	Script	Slow class	Flag
opening	4:00	3:40	4:16	ok
remember	6:00	4:36	5:28	ok
sneak	4:00	3:33	4:10	ok
explore	14:00	11:13	13:21	ok
mini	10:00	8:38	10:11	ok
break	4:00	3:28	4:07	ok

work	28:00	25:25	30:22	ok
talk	6:00	5:12	6:12	ok
exit	6:00	4:19	5:07	ok
look	3:00	2:28	2:55	ok
flex	5:00	0:38	0:44	ok
Total	90:00	73:10	86:54	

Assumptions as docs/audit/timing_p1.md: speech 120 words a minute, read-aloud 110, slide change 25 s, a group question 10 s thinking plus 20 s answers, papers to the camera 30 s. Workbook: Start here has 6 questions; a middle child needs about 7.5 min for page 1 and 7.0 min for page 2 (a slower child about 1.7 times as long, so may finish page 1 and part of page 2; that is fine, the check-together slides cover every answer). The lesson has no breakout rooms, so no breakout time is counted.